

984V/ 3.0mΩ·cm² Normally-off and 817V/ 2.7mΩ·cm² Normally-on GaN Trench CAVETs With Wrap-Around Gate

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Abstract—Gallium Nitride (GaN) trench Current Aperture Vertical Electron Transistors with wrap-around gate (WG-CAVETs) were demonstrated on GaN substrates. The periodic trench-groove structure enhanced gate control and suppressed gate leakage to $\sim 10^{-5}$ A/cm² up to 40 V gate bias. Reducing the groove spacing (S) increased threshold voltage (V_{TH}) and improved breakdown voltage. Both normally-on and normally-off WG-CAVETs were realized by varying S . A normally-on device with $S = 4$ μm achieved a specific on-resistance ($R_{ON,SP}$) of 2.7 mΩ·cm² and a breakdown voltage of 817 V. Reducing S to 2 μm enabled a normally-off device ($V_{TH} = 4.6$ V) with I_{ON}/I_{OFF} of 1.7×10^{10} , $R_{ON,SP}$ of 3.0 mΩ·cm², and 984 V breakdown. Also, a normally-off multi-finger WG-CAVET ($S = 2$ μm) delivered 0.87 A, with $R_{ON,SP}$ of 3.4 mΩ·cm² and a breakdown voltage of 597 V. For comparative analysis, WG-CAVETs were fabricated with a thinner regrown channel, decreasing from 140 nm to 50 nm. Normally-off devices were achieved over a broader range of groove spacings, albeit with a cost of degraded output current and breakdown voltage.

Index Terms—Gallium Nitride (GaN), Current Aperture Vertical Electron Transistor (CAVET), normally-off transistor, threshold voltage, vertical transistor.

I. INTRODUCTION

GALLIUM Nitride (GaN) High Electron Mobility Transistors (HEMTs) offer exceptional high-frequency switching and power handling capabilities, attributed to the wide bandgap of GaN and two-dimensional electron gas (2DEG). Vertical device topology is preferred for power electronics due to efficient footprint utilization, mitigated dispersion, avalanche capability, and superior thermal performance [1]. Among GaN vertical FETs, the Current

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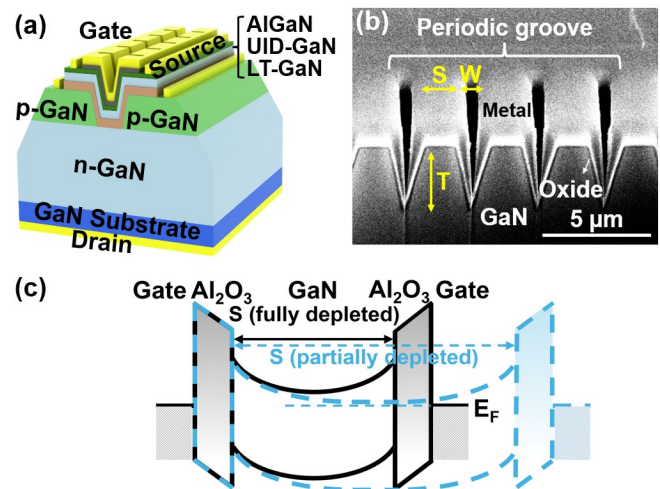


Figure 1. (a) Schematic of WG-CAVET. (b) Cross-section of periodic groove by FIB-SEM. (c) Schematic band diagram of groove region.

Aperture Vertical Electron Transistor (CAVET) [2-5] uniquely leverages a 2DEG channel for current conduction while blocking voltage through a vertical p-n junction [27,28]. This powerful combination makes CAVETs promising in low-to-medium voltage regimes, where the channel resistance dominates the total on-resistance [6].

Normally-off transistors with a high $V_{TH} > 3$ V and wide gate voltage swing are generally preferred for fail-safe switching and gate driver compatibility [7]. Various methods, including fluorine plasma [3] and p-GaN gate [8], have been used to achieve E-mode vertical GaN transistors with a 2DEG channel. However, despite these efforts, achieving a higher V_{TH} can come with the cost of 2DEG charge density reduction [9], which causes current degradation and high specific on-resistance ($R_{ON,SP}$).

We previously demonstrated the first normally-off GaN trench CAVET featuring a wrap-around gate [10]. Compared to conventional trench CAVETs, the WG-CAVET achieved $V_{TH} > 3$ V and improved breakdown voltage from 200 V to 600 V. Nevertheless, a relatively high off-state leakage current of the device remained an issue. In this work, we enhance the off-state blocking capability of WG-CAVETs by using a low-power post-groove etching process and increasing the drift layer thickness, successfully decreasing off-state leakage below 1 μA and achieving a higher breakdown voltage of 984 V. Additionally, by optimizing groove spacing (S) for devices with different channel thicknesses (t_{ch}), we demonstrate both

normally-on and normally-off WG-CAVETs.

II. DEVICE FABRICATION

Fig. 1 (a) illustrates the schematic of the wrap-around-gate CAVET (WG-CAVET). A periodic groove structure is etched into the trench region, enabling the gate electrode to wrap around both the top surface and the groove sidewalls. This structure is confirmed by the focused ion beam scanning electron microscopy (FIB-SEM) image in Fig. 1(b).

The GaN epitaxial structure was grown by metal-organic chemical vapor deposition (MOCVD), including an 8 μm n-type drift layer (Si: $1 \times 10^{16} \text{ cm}^{-3}$) and a 450 nm p-GaN (Mg: $1 \times 10^{18} \text{ cm}^{-3}$) on top. 600 nm deep apertures were etched on p-GaN. Then, UV/O₃ and HF dip were performed before regrowth to clean the surface [11]. The MOCVD regrowth included 100 nm low-temperature (LT) GaN to mitigate Mg out-diffusion [12, 13], a 140 nm (t_{ch}) UID-GaN channel, and 18 nm Al_{0.3}Ga_{0.7}N barrier. The device isolation included a mesa edge on the channel and a 5-degree beveled edge on the p-n junction [14]. The buried p-GaN was activated by rapid thermal annealing (RTA) in N₂ at 900 °C for 10 min. 20/120/30/50 nm Ti/Al/Ni/Au source electrodes were deposited and annealed via RTA at 850 °C for 45 sec. A 3 μm deep periodic groove structure was etched using inductively coupled plasma-reactive ion etching (ICP-RIE), including a deep GaN etching of 2.8 μm at 25 W RF power, followed by a post-etch of 200 nm at 5 W RF power to reduce surface damage near the groove bottom. After UV/O₃ cleaning, 100 nm Al₂O₃ was deposited by ALD, followed by post-deposition annealing by RTA at 350 °C for 10 min. 30/200 nm Ni/Au gate was deposited on top, and 30/150 nm Ti/Au drain electrode was deposited on the wafer backside.

III. DEVICE RESULTS AND DISCUSSION

Fig. 2 (a) shows the measured I_D - V_{GS} transfer characteristics of WG-CAVET with different S . Compared to our prior work

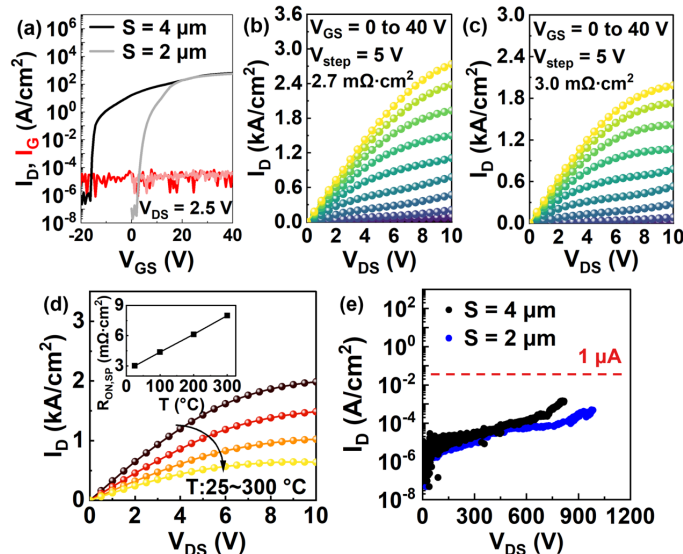


Figure 2. (a) I_D - V_{GS} measurements of WG-CAVET with $S = 4 \mu\text{m}$ and $S = 2 \mu\text{m}$. I_D - V_{DS} measurements of the device with (b) $S = 4 \mu\text{m}$ and (c) $S = 2 \mu\text{m}$. (d) Temperature-dependent $R_{\text{ON,SP}}$ of WG-CAVET with $S = 2 \mu\text{m}$ extracted from I_D - V_{DS} measurements at $V_{GS} = 40 \text{ V}$. (e) Measured off-state characteristics with different S .

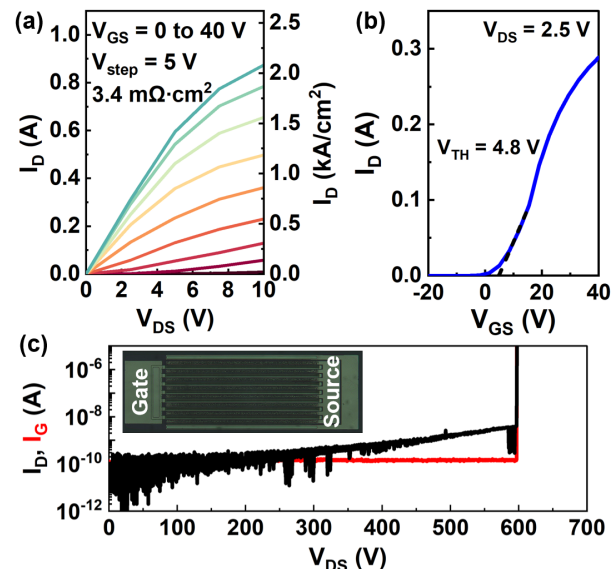


Figure 3. (a) I_D - V_{DS} measurement and (b) I_D - V_{GS} measurement of large-area WG-CAVET with $S = 2 \mu\text{m}$. (c) Optic microscope image of the device in the inset, and measured off-state characteristics show a breakdown voltage of 597 V.

[10], the groove width W was set to $1 \mu\text{m}$ in this work to retain more 2DEG channel. For $S = 4 \mu\text{m}$, the extracted V_{TH} was -15.2 V at $I_D = 1 \mu\text{A/mm}$, with $I_{\text{on}}/I_{\text{off}}$ of 1.6×10^9 and gate leakage $\sim 10^{-5} \text{ A/cm}^2$ at $V_{GS} = 40 \text{ V}$. Reducing S to $2 \mu\text{m}$ enabled normally-off operation, yielding $V_{\text{TH}} = 4.6 \text{ V}$, $I_{\text{on}}/I_{\text{off}} = 1.7 \times 10^{10}$, and similarly low gate leakage. Fig. 2 (b) and (c) illustrate the measured I_D - V_{DS} characteristics for $S = 4 \mu\text{m}$ and $2 \mu\text{m}$, with $R_{\text{ON,SP}}$ of $2.7 \text{ m}\Omega\cdot\text{cm}^2$ and $3.0 \text{ m}\Omega\cdot\text{cm}^2$, respectively. The positive V_{TH} shift with smaller S comes at the cost of a slight increase in $R_{\text{ON,SP}}$ due to reduced 2DEG channel area. Reducing S beyond this could increase V_{TH} further, but it would compromise output current by excessive 2DEG area reduction.

Elevated-temperature characterization of WG-CAVET with $S = 2 \mu\text{m}$ is shown in Fig. 2 (d). $R_{\text{ON,SP}}$ from I_D - V_{DS} measurements rises to $8.0 \text{ m}\Omega\cdot\text{cm}^2$ at $300 \text{ }^\circ\text{C}$, primarily limited by optical phonon scattering [29]. Fig. 2 (e) shows the off-state characteristics. The device with $S = 4 \mu\text{m}$ exhibited a measured breakdown voltage of 817 V , while reducing S to $2 \mu\text{m}$ increased it to 984 V . Both devices kept off-state leakage current $< 1 \mu\text{A}$ before breakdown. Compared to conventional gate structures that only modulate from the top surface of AlGaIn/GaN, WG-CAVETs apply gate control from both the top surface and the vertical groove sidewalls. As S decreases, the depletion regions under the gate extending from the groove sidewalls overlap more effectively in the AlGaIn/GaN regions, facilitating full pinch-off and suppressing off-state leakage. Smaller S reduces field crowding near groove corners, resulting in higher breakdown voltage [10].

A large-area WG-CAVET with seven gate fingers was fabricated, with an active area of 0.042 mm^2 . The groove dimensions were $W = 1 \mu\text{m}$, $T = 3 \mu\text{m}$, and $S = 2 \mu\text{m}$. The output current reached 0.87 A from I_D - V_{DS} measurements by using this multi-finger structure (Fig. 3 (a)). The extracted on-resistance was 8.06Ω ($R_{\text{ON,SP}} = 3.4 \text{ m}\Omega\cdot\text{cm}^2$). The I_D - V_{GS} measurement in Fig. 3 (b) shows a V_{TH} of 4.8 V by linear extrapolation. The measured breakdown voltage (Fig. 3 (c))

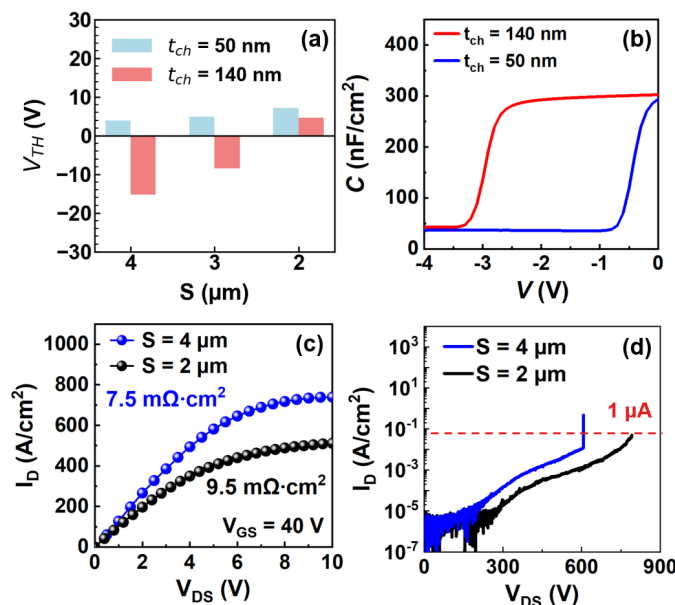


Figure 4. (a) Measured V_{TH} comparison of devices with different t_{ch} and different S . (b) C - V measurements of regrown AlGaIn/GaN on CAVET samples with different t_{ch} . (c) I_D - V_{DS} measurements at $V_{GS} = 40$ V and (d) measured breakdown voltage of WG-CAVETs ($t_{ch} = 50$ nm) with different S .

was 597 V, limited by localized dielectric breakdown.

Fig. 4 (a) summarizes the measured V_{TH} with different S . For devices with a channel thickness (t_{ch}) of 140 nm, V_{TH} increases from -15.2 V to -8.4 V as S reduces from 4 μm to 3 μm , and it shifts to $+4.6$ V when $S = 2$ μm . Since the periodic groove structure is etched into the AlGaIn/GaN region, the groove spacing S corresponds to the remaining AlGaIn/GaN channel width between adjacent grooves. As S decreases, the wrap-around gate sections facing each other across this channel become more closely spaced, which yields more effective channel control. In the horizontal cross section (Fig. 1(c)), the conduction band bending originating at each groove sidewall extends farther inward as S is reduced. Once the resulting depletion regions overlap, the channel is pinched off, causing a positive shift in V_{TH} . WG-CAVETs with a thinner t_{ch} of 50 nm were fabricated as a comparison, enabling normally-off operation with greater tolerance for variations in S . Devices achieved measured $V_{TH} = 3.9$ V for $S = 4$ μm and $V_{TH} = 7.2$ V for $S = 2$ μm . However, 2DEG density of the regrown AlGaIn/GaN channel was reduced from 5.6×10^{12}

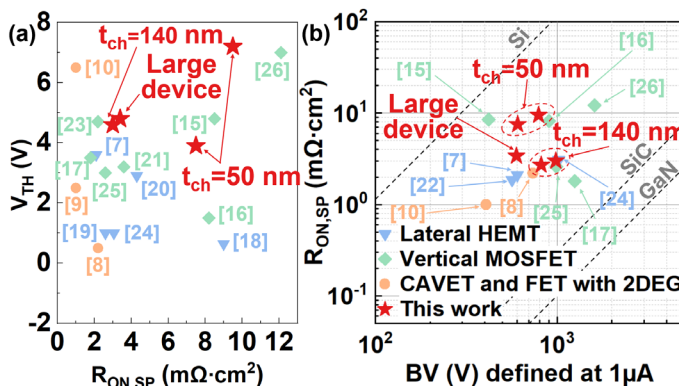


Figure 5. Performance comparison of WG-CAVETs in this work with lateral and vertical GaN power transistors reported.

cm^{-2} to $1.5 \times 10^{12} \text{ cm}^{-2}$ by reducing the t_{ch} from 140 nm to 50 nm, as extracted from C - V measurements at 1 MHz in Fig. 4 (b). The reduced separation between the 2DEG and the regrowth interface leads to a narrower current path and increases compensation effects due to Mg redistribution into the channel during regrowth [12]. As t_{ch} decreases to 50 nm, the $R_{ON,SP}$ extracted from I_D - V_{DS} measurements increased to $7.5 \text{ m}\Omega \cdot \text{cm}^2$ for $S = 4$ μm and $9.5 \text{ m}\Omega \cdot \text{cm}^2$ for $S = 2$ μm (Fig. 4 (c)). Measured breakdown voltage decreased to 607 V for $S = 4$ μm and 791 V for $S = 2$ μm (Fig. 4 (d)). The lower breakdown for devices with $t_{ch} = 50$ nm compared to 140 nm may result from a higher electric field under the gate due to the thinner channel.

Fig. 5 shows the performance of WG-CAVETs, compared with several reported GaN vertical and lateral FETs, including V_{TH} vs. $R_{ON,SP}$ and $R_{ON,SP}$ vs. breakdown voltage (BV) defined at a threshold current level of $1 \mu\text{A}$ [7-10, 15-26]. Compared with our prior report [10], this work achieves a more balanced performance among V_{TH} , $R_{ON,SP}$, and breakdown voltage in CAVETs, improving off-state block capability while keeping robust normally-off operation.

IV. CONCLUSION

Both normally-on and normally-off GaN WG-CAVETs were demonstrated. A combination of medium and low power etching techniques mitigated surface damage during groove formation. For $t_{ch} = 140$ nm, the normally-on WG-CAVET ($S = 4$ μm) achieved $R_{ON,SP}$ of $2.7 \text{ m}\Omega \cdot \text{cm}^2$ and a breakdown voltage of 817 V. The normally-off device ($S = 2$ μm) exhibited V_{TH} of 4.6 V, $R_{ON,SP}$ of $3.0 \text{ m}\Omega \cdot \text{cm}^2$, and a breakdown voltage of 984 V. Furthermore, a multi-finger normally-off WG-CAVET achieved a high output current of 0.87 A. Reducing t_{ch} to 50 nm accommodated a wider range of S but at a cost of lower output current and earlier breakdown.

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